AE 358 RENEWABLE ENERGY SYSTEMS AND MANAGEMENT

LECTURE 7

SOLAR ENERGY

Lecturer: Francis Kemausuor

Lecture outline

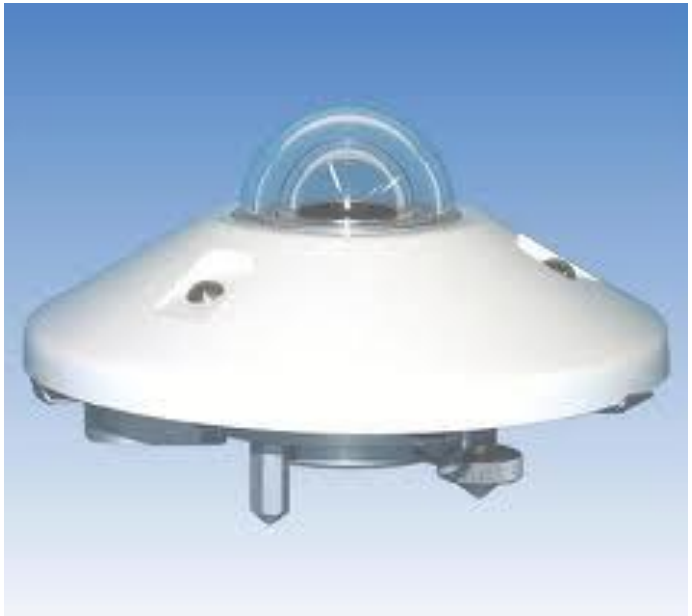
- Explain how energy from the sun is converted to electricity and heat using solar photovoltaic and solar thermal applications
- Discuss the applications of solar photovoltaics

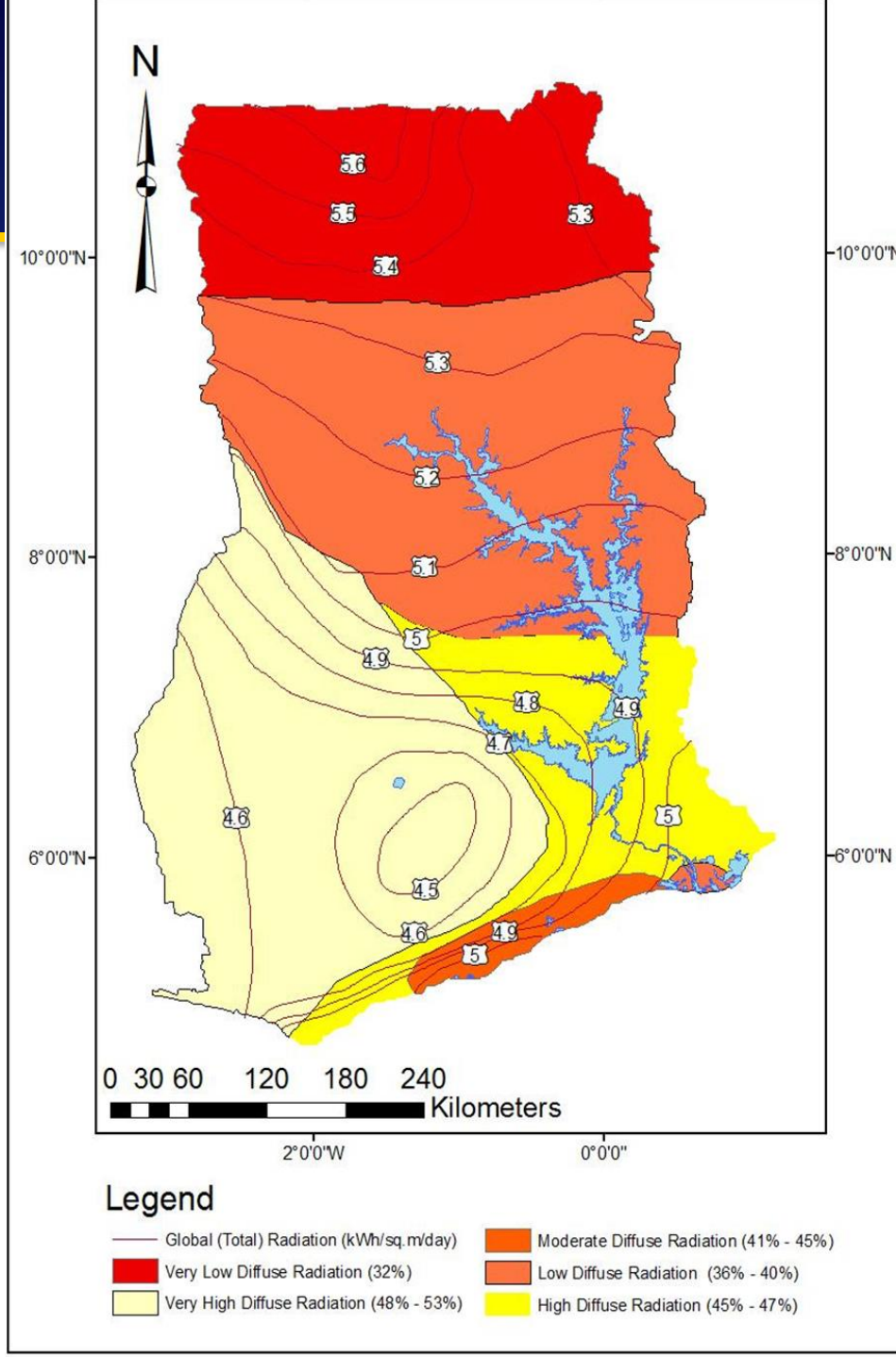
Solar radiation

- The radiation coming directly from the sun and received at the earth's surface on a unit area normal to the solar radiation is called direct solar radiation.
- Diffuse solar radiation is solar radiation reaching the Earth's surface after having been scattered from the direct solar beam by molecules or suspensoids in the atmosphere.
- Insolation is a measure of solar radiation energy received on a given surface area in a given time.

Solar radiation

- The measurements are taken using either a pyranometer (measuring global radiation) and/or a pyrheliometer (measuring direct radiation) and the unit is kW/m^2 or kWh/m^2 per day



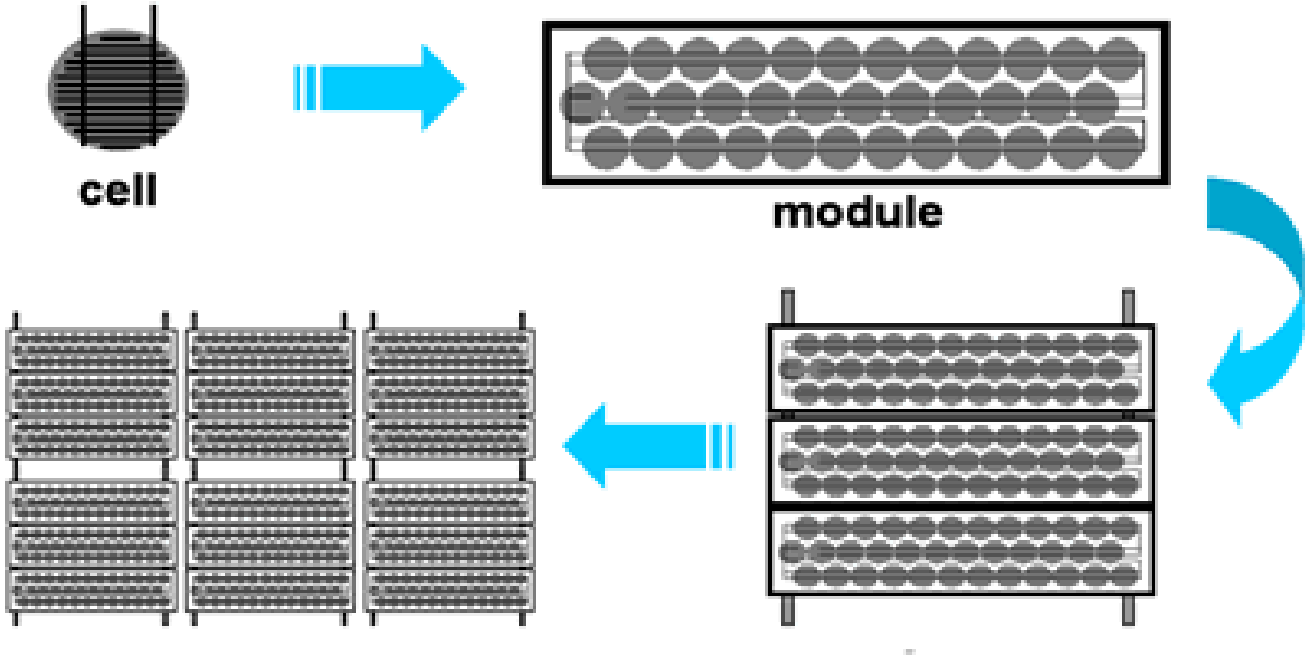


Solar energy technologies

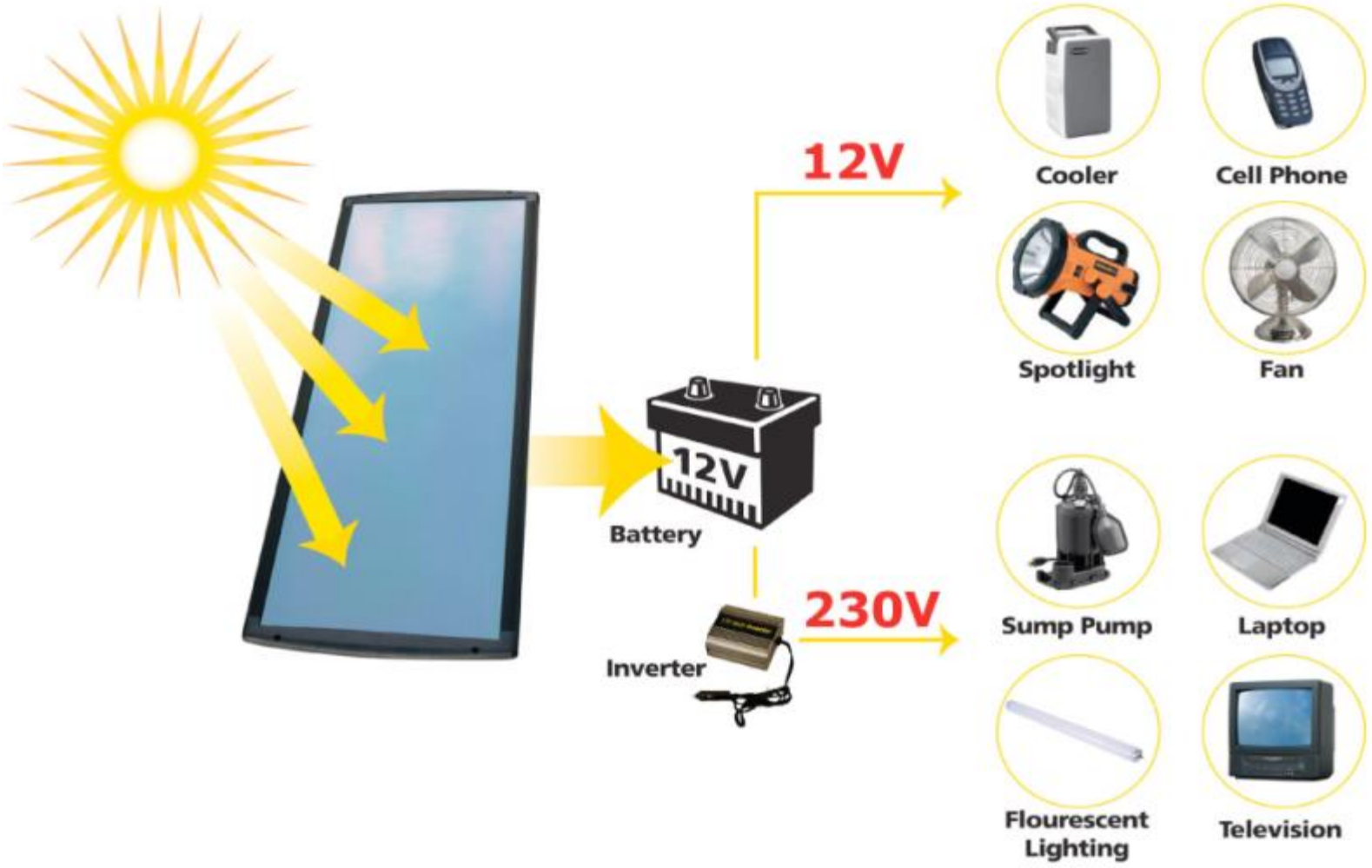
- Two main categories:
  - Solar photovoltaic
  - Solar thermal
- Solar thermal systems use the sunlight to produce directly usable thermal energy or heat for water or space heating, drying, cooking, cooling or electricity generation while solar photovoltaic is the direct conversion of solar radiation into electricity.

Photovoltaic (PV) systems

- Photovoltaic is the field of technology and research related to the devices which directly convert sunlight into electricity.

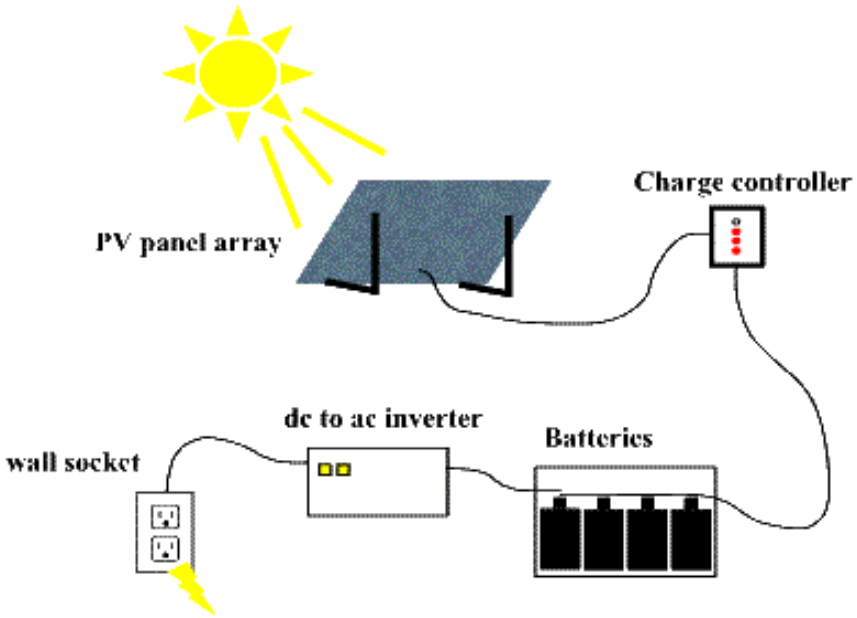


Application of Solar Panels



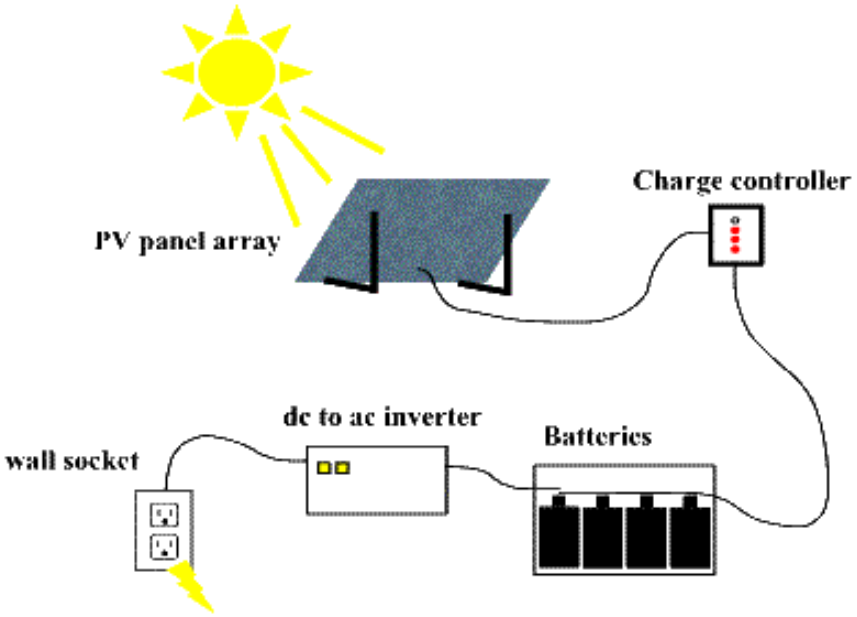
Systems and Components

- **The Generator**
  - The heart of a photovoltaic system is the generator itself. The generator is the connection of any number of solar modules (also called panels).
  - Solar modules are rated in peak watts (Wp) according to their output under optimal outdoor conditions. Thus, a 50 Wp module can be expected to supply 50 W of power in full sunshine.



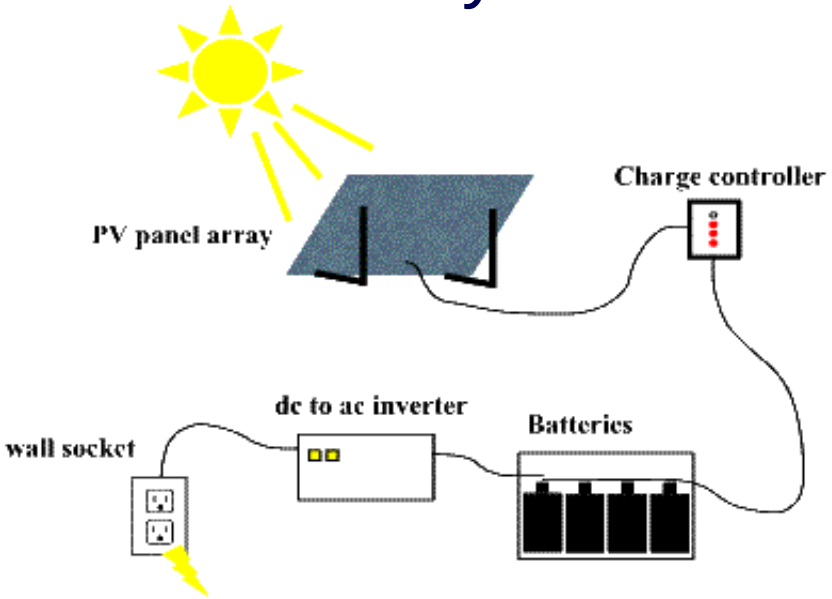
Systems and Components

- **The battery**
- Only lead-acid batteries: Automotive batteries and Solar batteries



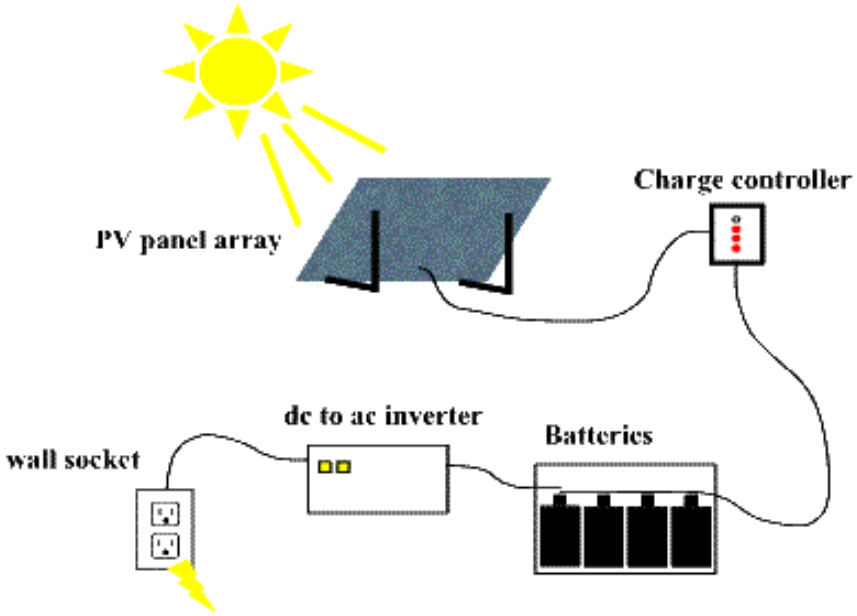
Systems and Components

- **The controller**
- Protects battery against overcharging and discharging
- Charge controllers are also referred to as charge regulators and battery control units (BCU).



Systems and Components

- **The inverter**
- An inverter transforms low-voltage DC supplied by a solar system into high-voltage AC. The input of an inverter is designed for 12 V (24 V, 48 V, etc.), depending on the type. At the output it produces 110 or 240 V AC.



Sizing the PV system

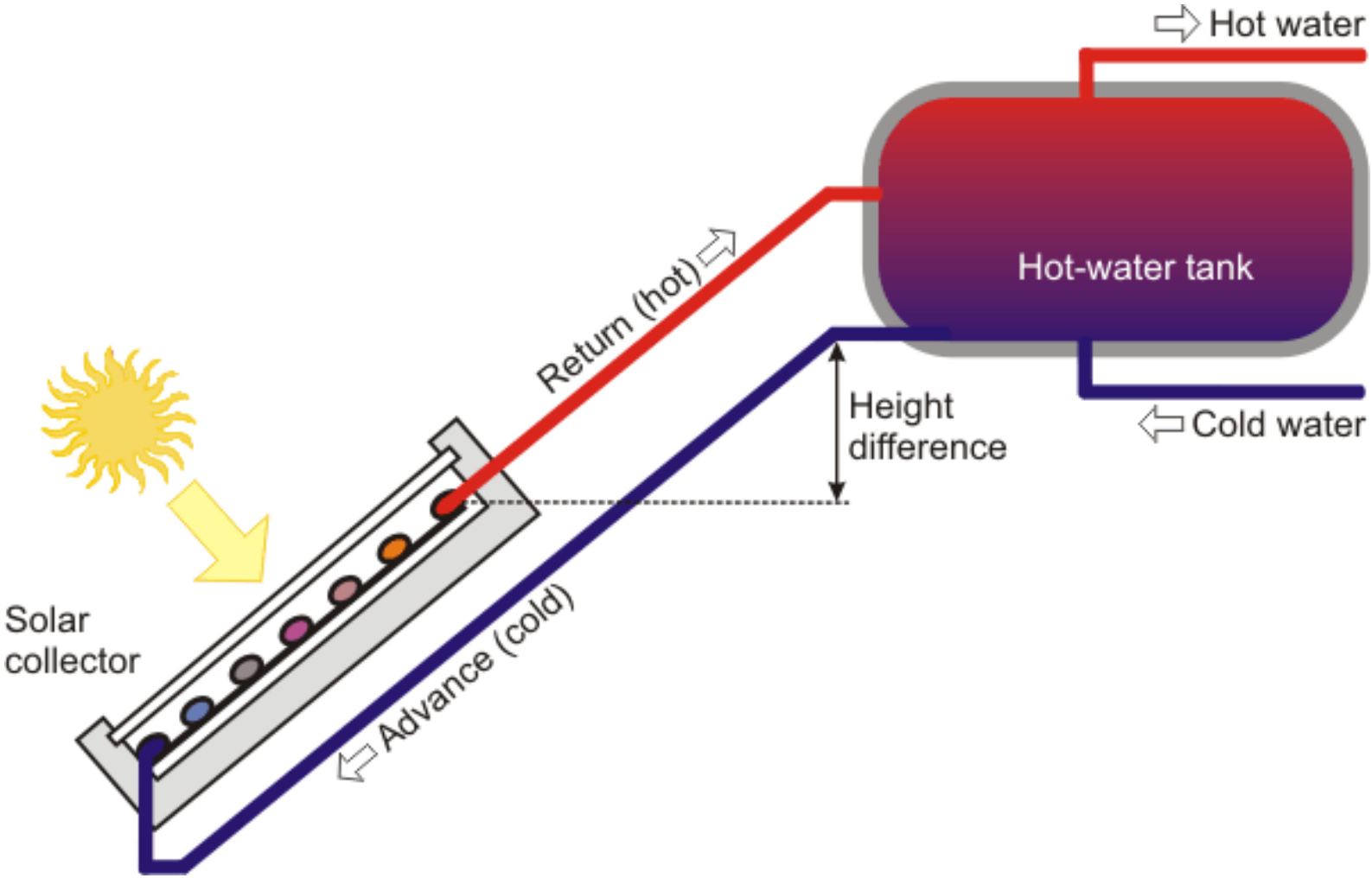
- System sizing means adapting the supply to the demand.
  - Calculating the demand
  - Sizing the generator
- The generator has to cover the energy consumption (ELOAD) plus the system losses. The system losses are calculated by the efficiencies (η) of the components. The Energy required,

$$E_{GEN} \approx \frac{E_{LOAD}}{0.85 \times \eta_{BAT} \times \eta_{INV}}$$

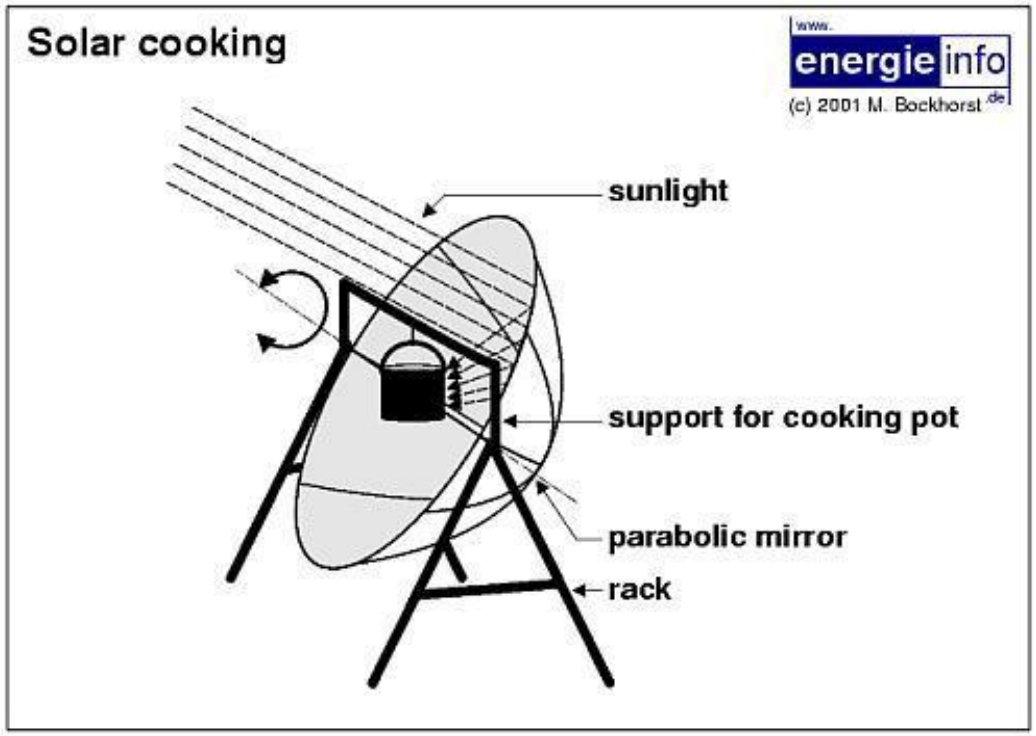
Solar thermal applications

- Solar Thermal Collectors convert solar radiation into heat.
- Solar thermal applications include:
  - Solar Water Heating
  - Solar Cooking
  - Solar Drying
  - Solar Water Distillation, etc.

Solar water heater



Solar cooker





QUESTIONS